

Assignment - 3

```
// level 1
// 1. Create array
let fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"];
console.log("1:", fruits);

// 2. First & Last element
console.log("2:", fruits[0], fruits[fruits.length - 1]);

// 3. Length
console.log("3:", fruits.length);

// 4. push()
fruits.push("Pineapple");
console.log("4:", fruits);

// 5. pop()
fruits.pop();
console.log("5:", fruits);

// 6. unshift()
fruits.unshift("Strawberry");
console.log("6:", fruits);

// 7. shift()
fruits.shift();
console.log("7:", fruits);

// 9. Combine arrays
let arr1 = [1, 2];
let arr2 = [3, 4];
let combined = [...arr1, ...arr2];
console.log("9:", combined);

// 10. includes()
console.log("10:", fruits.includes("Mango"));

// level2

let numbers = [10, 20, 30, 40, 50];
```

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37
38 // level2
39
40 let numbers = [10, 20, 30, 40, 50];
41
42 // 11. for loop |
43 console.log("11:");
44 for (let i = 0; i < numbers.length; i++) {
45 |   console.log(numbers[i]);
46 | }
47
48 // 12. for...of
49 console.log("12:");
50 for (let num of numbers) {
51 |   console.log(num);
52 | }
53
54 // 13. forEach()
55 console.log("13:");
56 numbers.forEach((val, index) => {
57 |   console.log(index, val);
58 | });
59
60 // 14. Sum
61 let sum = 0;
62 for (let num of numbers) {
63 |   sum += num;
64 | }
65 console.log("14:", sum);
66
67 // 15. Max
68 let max = numbers[0];
69
```

```
0 // 16. Count even
1 let evenCount = 0;
2 for (let num of numbers) {
3   |   if (num % 2 === 0) evenCount++;
4 }
5 console.log("16:", evenCount);
6
7 // 17. Pass/Fail
8 let markss = [30, 40, 80, 20];
9 console.log("17:");
0 marks.forEach(m => {
1   |   console.log(m, m >= 35 ? "Pass" : "Fail");
2 });
3
4 // 18. Names starting with A
5 let names1 = ["Arun", "Bala", "Anu", "Kumar"];
6 console.log("18:");
7 names.forEach(n => {
8   |   if (n.startsWith("A")) console.log(n);
9 });
0
1 // 19. Reverse
2 let rev = [...numbers].reverse();
3 console.log("19:", rev);
4
5 // 20. Sort ascending
6 let sorted = [...numbers].sort((a, b) => a - b);
7 console.log("20:", sorted);
```

js-special > day3 > Assignment > JS day3.js > sum1

70 // Base array

71 let nums = [10, 25, 60, 80, 120];

72

73 // 21. Double each element

74 let doubled = nums.map(n => n * 2);

75 console.log("21:", doubled);

76

77 // 22. Only even numbers

78 let even = nums.filter(n => n % 2 === 0);

79 console.log("22:", even);

80

81 // 23. Total sum

82 let sum = nums.reduce((acc, n) => acc + n, 0);

83 console.log("23:", sum);

84

85 // 24. First number > 50

86 let firstBig = nums.find(n => n > 50);

87 console.log("24:", firstBig);

88

89 // 25. Any number > 100

90 let check100 = nums.some(n => n > 100);

91 console.log("25:", check100);

92

93 // 26. All numbers positive

94 let allPositive = nums.every(n => n > 0);

95 console.log("26:", allPositive);

96

97 // 27. Names to uppercase

98 let names = ["selin", "john", "arun"];

99 let upperNames = names.map(n => n.toUpperCase());

100 console.log("27:", upperNames);

101

102 // 28. Marks >= 50

103 let marks = [40, 55, 70, 30, 90];

104 let pass = marks.filter(m => m >= 50);

105 console.log("28:", pass);

106

107 // 29. Multiply all by 10

108 let times10 = nums.map(n => n * 10);

109 console.log("29:", times10);

110

111 // 30. Average using reduce

112 let avg = nums.reduce((acc, n) => acc + n, 0) / nums.length;

113 console.log("30:", avg);

114